

Review

Phylogenomics beyond the
'bag-of-genes' modelIker Rivas-González ¹, Siavash Mirarab ^{2,*}, and Edward L. Braun ³

Genome-scale data have transformed phylogenetics, yet most phylogenomic analyses disregard the genomic positions of the loci used to infer evolutionary histories. Although the role of deep coalescence is well studied, phylogenomics has paid far less attention to the fact that recombination generates correlated genealogies, with positional dependencies encoding rich information about demography, selection, and gene flow. Recent advances in genome assembly and computational modeling now make it feasible to move beyond long-standing simplifications that treat genomes as 'bags of genes'. Here, we synthesize the growing literature on position-aware phylogenomic approaches, from site-based methods to coalescent hidden Markov models. We argue that explicitly incorporating genomic position reveals evolutionary signals inaccessible to traditional approaches and represents a necessary step toward a more realistic framework for genome evolution.

Position-aware phylogenomics: why and why now?

Phylogenomics (Box 1) is now the standard framework for studying evolutionary history. Initially, there was optimism that genome-wide data would yield trees with 'unequivocal support' [4], but subsequent studies have shown that the exuberant pursuit of fully resolved species trees is unjustified [5]. The basis for this change in attitude was the observation that the histories of different genomic regions can be highly divergent [6]. Although some observed incongruence between genomic regions is a methodological mirage [7–9], much of it reflects real heterogeneity in evolutionary histories. The heterogeneity itself is interesting, as its levels and how it changes across the genome directly relate to parameters of interest, such as ancestral population sizes, amounts of gene flow among species, and shifts in the tempo of evolution. Moreover, we have statistical models of heterogeneity derived from coalescence theory [3], and identifying regions that do not conform to these models provides an opportunity to discover evidence for more complex processes, such as introgression and selection (e.g., [10–16]). The real heterogeneity in evolutionary history is attributed to two phenomena: **incomplete lineage sorting (ILS)** (see Glossary) and reticulate evolution (see Box 1 for caveats). These two phenomena, which are not exclusive, are fundamentally related to recombination [17], which creates breaks in ancestry along the genome (Figure 1A). Despite this centrality of recombination, the literature, with its wealth of methods [18–20], has mostly sought ways to avoid directly modeling recombination and, more broadly, the positions of loci along the genome. Instead, a 'bag-of-genes' model that treats loci as independent entities has been adopted.

The growing availability of very high-quality genome assemblies [21] will provide accurate positional information for loci used in phylogenetic inference. This rich positional information has the potential to give insights into evolutionary history that go beyond the species tree topology. For example, since introgression interacts with recombination in complex ways, positional information can help clarify the recombinational landscape and thus improve introgression detection (a topic recently reviewed by Burbrink *et al.* [22]). Although the widespread use of position-aware phylogenomics presents many challenges, the increased computational power now available provides an

Highlights

Phylogenomics has mostly adopted a bag-of-genes model, treating loci as recombination-free but fully independent.

Evidence is accumulating that accounting for genomic positions can improve species tree inference and facilitate the detection of changes in evolutionary parameters such as population size, recombination rates, and gene flow.

Methods that go beyond the bag-of-genes model have been developed, including those that work directly on sites, examine gene trees as sequences *post hoc*, or use hidden Markov models to model locational dependency.

With the advent of new methods and higher quality genomes, the time is ripe for a paradigm shift in phylogenomics to account for genomic position in all analyses.

The time is also ripe for much-needed mathematical, algorithmic, and conceptual advances that enrich the toolkit available for such position-aware phylogenomic analyses.

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Box 1. Key concepts referenced in phylogenomics literature

Phylogenomics: The inference of phylogenetic trees from genome-wide data.

Coalescent process: Tracing ancestries back in time through generations leads to finding common ancestors, called coalescence. Kingman's process is used to model coalescence [1]. Recombinations create discontinuities in the coalescent history, making it unique to each genomic region (Figure 1).

Species tree: A tree that shows the history of speciation events.

MSC: An extension of Kingman's coalescent to a species tree, allowing independent processes on each branch, and moving uncoalesced lineages up to parent species.

Coalescence gene (c-gene [2]): A region of the genome where all nucleotides have the same genealogy, unbroken by recombination events that result in different tree topologies or coalescent times. The boundaries of c-genes depend on the sampled individuals/species.

Gene tree (locus tree, local tree, and genealogy): Here, the evolutionary history of a single c-gene. Nodes of gene trees represent coalescent events. The literature uses 'gene trees' more broadly for trees inferred from homologous regions of the genome, whether or not they are c-genes.

Orthologs and paralogs: Duplication of a region of the genome creates paralogous copies, in contrast to orthologs, which correspond to the same (ancestral) copy. Gene duplication and loss (GDL) can create multi-copy gene trees with paralogous copies at leaves and some internal nodes corresponding to gene duplication events. Mistaken (hidden) paralogy confuses coalescence and duplication events.

ILS: When tracing the lineage from two branches of a species tree, they may first coalesce in their last common ancestor; 'deep' coalescence is when they do not and instead coalesce in branches further back in time. With deep coalescence, the genealogy may not match the species tree topology [3], resulting in ILS. ILS is possible for any species tree but has high probability when population size is large relative to the number of generations between speciation events.

Reticulate evolution: Speciations through hybridization and gene flow between species break the tree-like structure of species histories. Similarly, HGT does not follow a species tree model. Hybridization is assumed rare, while HGT is assumed common only among prokaryotes.

Discordance (incongruence): Differences among gene trees inferred from sequence data can originate from biological causes such as ILS, hybridization, HGT, and GDL or artifacts such as inference uncertainty, systematic bias, undetected recombinations, undetected paralogy, and alignment errors.

Glossary

Ancestral recombination graph

(ARG): the most detailed model of genome evolution, showing both coalescent and recombination events (Figure 1).

Hidden Markov model (HMM): a graphical model consisting of a hidden Markov chain and observed data for each hidden element.

Incomplete lineage sorting (ILS): resulting in gene tree discordance due to stochastic coalescence processes (Box 1).

Multi-species coalescent (MSC): the MSC model is the main model used to study ILS.

Sequentially Markovian coalescent (SMC): a simplification of the recombination-with-coalescent process that assumes gene trees are Markovian.

opportunity to implement more advanced modeling tools. Here, we synthesize the recent literature on the use of positional information in phylogenomics, focusing on modeling and methodological questions, and look forward to a time when incorporating this information is routine.

Impacts of recombination from individuals to species

Recombination events, which originate at the individual level by breaking the ancestry into separate lineages for different parts of the chromosome (Figure 1A), can (but do not always) leave a mark at phylogenetic scales (Figure 1B,C). Across individuals within a population, the lineage trees of recombination-free regions can differ from those of other regions due to distinct ancestries. In a backward-in-time coalescence model, each recombination event creates an opportunity for the two sides of the breakpoint to have different coalescent histories [23]. The 'coalescent-with-recombination' statistical model captures this dynamic, allowing lineages to either coalesce or break moving back in time. Originally conceived as a temporal process where lineages merge (coalescent events) or split (recombination events) backward in time [24–26], the coalescent-with-recombination model can also be understood as a spatially sequential process [27], where a recombination event might change branch lengths and topology of a tree as we move along the genome. In this model, neighboring lineage trees are not identical, but they are not independent either, since they must be the same except for the parent chosen at a single recombination breakpoint. The resulting set of correlated gene trees along the genome

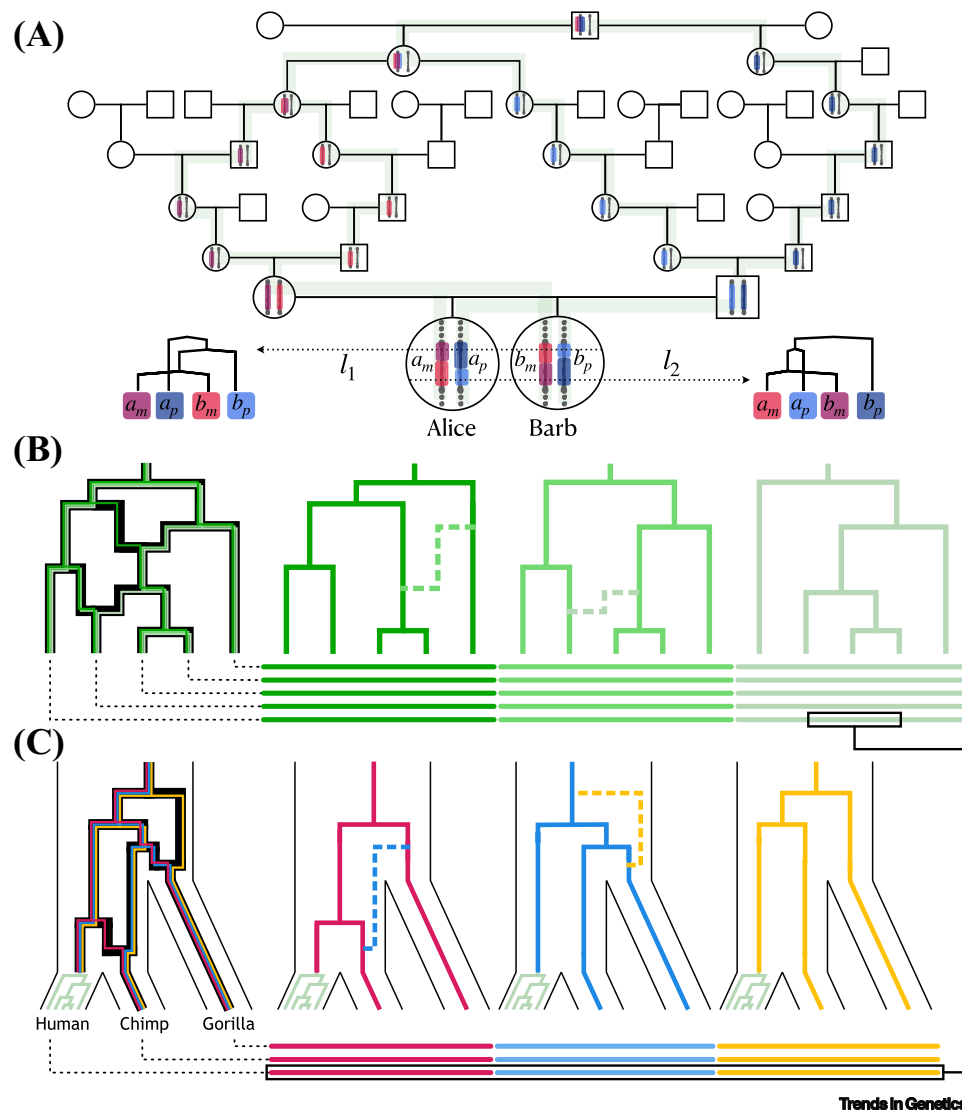


Figure 1. Recombinations create mosaics of histories from individuals to species. (A) At the individual level, recombinations create mosaics of histories. Here, tracing back in time the relationships between the maternal and paternal alleles of two sisters, Alice and Barb (a_m, a_p, b_m, b_p), leads to different lineage trees depending on whether locus l_1 or locus l_2 is examined. For simplicity of exposition, we show a region that has recombined in both Alice and Barb but not in any of the prior seven generations shown. Coalescences correspond to two traced lineages reaching the same ancestor from different paths. (B) The same process results in an ancestral recombination graph (ARG) across multiple individuals in a population. Each recombination event prunes a subtree and regrafts it at an earlier time in the lineage tree (dashed lines); some such events change the tree (topology or length), creating a new coalescence gene (three shown, in shades of green). (C) The ARGs can then be extended to multiple species, imposing further constraints on where coalescences and regrafts can happen.

(Figure 1B) is collectively known as the **ancestral recombination graph (ARG)**, which records where in the genome and when in the ancestral history of the samples recombination and coalescent events occurred (Box 2).

The recombination-with-coalescence model can be extended to multiple populations and/or species, guided by a species tree. The same process occurs on each ancestral branch of the species tree, and lineages from descendants continue coalescing within their parent populations

Box 2. Species tree inference under MSC: algorithmic concepts

Statistical consistency: A statistical estimator is consistent if it converges to the true value of the estimated parameter (e.g., the species tree topology) as the sample size grows without bound. A positively misleading estimator can converge on an incorrect parameter in expectation.

Summary methods: The scalable two-step approach for inferring species trees: (i) gene trees are estimated using alignments of individual loci and (ii) those gene trees are combined to yield the species tree.

Quartet trees: A tree can be induced down to any four taxa (quartet) by pruning other species and suppressing degree-two nodes. The resulting 'quartet trees' are the smallest trees with any topological information for unrooted trees.

Anomaly zones and the significance of quartets: The 'anomaly zone' is the part of species tree space where the gene tree topology with the highest probability differs from the species tree [28]. There is no anomaly zone for unrooted four-taxon species trees under the MSC [29] or several GDL and GDL + ILS models [30–32], a fact underlying many summary methods.

ASTRAL: The species tree that maximizes the total number of induced quartet trees shared with all gene trees is a statistically consistent estimator; ASTRAL efficiently finds such a 'median tree' using dynamic programming [33].

SPR: Topological rearrangement for phylogenetic trees that involves removing a group of taxa from one place in the tree and placing it elsewhere in the tree. Trees that differ due to a single recombination event differ by an SPR rearrangement (with some restrictions on the specific SPR moves allowed [34]).

Ultrametricity: A tree where all leaves have the same path length to the root is called ultrametric. Ultrametricity is expected for branch length in time units or when evolutionary parameters (rates and population sizes) are assumed constant. Population genetics mostly uses ultrametric trees, while phylogenetics does not.

ARG: A graph that represents the complete genealogical history of genomes along a recombining chromosome, with nodes that correspond to coalescent or recombination events, fully annotated with position information. The annotations enable recovery of 'local trees' along the genome, akin to gene trees defined in Box 1 and shown in Figure 1.

(Figure 1C). This model effectively enforces constraints that prevent coalescence after speciation events. Continuous gene flow can then be understood as exceptions to these constraints, and hybridization as alternative constraints. Thus, regardless of the cause of discordance (ILS or hybridization), recombination is the underlying force creating heterogeneity in the first place for sexually reproducing organisms. Among non-sexually reproducing organisms (e.g., prokaryotes), horizontal gene transfer (HGT) [35,36] is the main mechanism that creates heterogeneity; while we do not focus on HGT here, some forms of HGT (such as homologous recombination) can be similar to recombination in meiosis in terms of their effect on gene trees [37].

While ARGs encode full genealogical information, including recombination breakpoints and all gene trees, they are challenging to infer. In the full coalescence-with-recombination model [27], the process is not Markovian, as dependencies can be long range [38]. However, keeping track of genealogical information across all previous sites is computationally infeasible [38]. As a tractable approximation to the coalescent with recombination, the **sequentially Markovian coalescent (SMC)** was developed [38,39], in which the genealogy at a given site is assumed to depend only on the genealogy at the previous site. This approximation has been proven to be highly accurate and is used extensively in population genetics [40]. Because the SMC opened up the possibility of using standard **hidden Markov model (HMM)** machinery, inferring the ARG became computationally tractable. The SMC has been extensively used to infer changes in effective population size (N_e) and to reconstruct coalescent times for a set of samples within a single population—see [41–43]. Other methods, collectively known as coalescent hidden Markov models, add another layer of complexity by temporally restricting when lineages can coalesce. We return to these in more detail below. Moving away from SMC, recent advances in ARG reconstruction methods (such as posterior re-sampling, threading, and other mathematical approximations) have enabled reconstruction of the genealogical history of hundreds or even thousands of samples (see ARGweaver [44], Relate [45],

tsinfer + tsdate [46,47], or ARG-Needle [48]). In contrast to the bag-of-genes model adopted in phylogenomics, these methods provide a rich output with positional information along the genome.

Despite their elegance, ARGs and their approximations have mostly been confined to population genetics because extending them to deeper time scales presents challenges. Although there have been attempts to infer natural selection (in the context of trans-species polymorphism [49]) and gene flow [50] using ARGs reconstructed from samples among populations or species, their applicability remains limited. ARG-based methods make a series of assumptions that are unrealistic at deep timescales and across multiple species. At the phylogenetic scale, the infinite-sites assumption breaks down (necessitating the use of more complex sequence-evolution models), and strict molecular clock assumptions are invalid (making trees nonultrametric; see Box 2). Inference under these conditions creates challenges—in particular, lack of ultrametricity—that make it harder to infer ARGs. Phylogenomic methods have mitigated these added complexities by adopting a simplifying independence assumption.

The independence assumption—aka the bag-of-genes model

The **multi-species coalescent (MSC)** literature has been dominated by the bag-of-genes model, which entails two ideas: First, it assumes that loci included in the analysis are far enough apart that recombination between them results in their having independent evolutionary histories. With this assumption, we can treat lineage trees as independent of one another, ignore their positions along the genome, and treat each gene tree as a draw from an underlying distribution. As a consequence, we lose spatially encoded genealogical information. Second, it assumes a single evolutionary history for each locus; in other words, it assumes that the loci used for analyses correspond to coalescence genes, without intralocus recombination events that change the tree topology or branch length (i.e., a c-gene; see Box 1). In practice, we often leave this assumption unexamined and assume that short genomic regions have few, if any, recombinations of consequence. Only recently have researchers begun to develop complex analytical models for characterizing c-gene length [51].

Given the obvious flaws, one may ask why the independence assumption was widely adopted. Two broad reasons contributed. First, data generation was insufficient both in quantity and in quality at the beginning of the phylogenomic era. High-quality genome assemblies were rare, and methods used to obtain sequence data, such as transcriptome and PCR-based amplicon sequencing, inherently lose information about genomic context. Even with whole-genome data, the simplest pipelines for finding orthology relied on predefined gene boundaries, which could be implemented by adopting annotation pipelines as opposed to more complex whole-genome alignment. Many of these pipelines have endured, perhaps because annotation remains easier than alternatives. Moreover, sequencing approaches that inherently lose information about genomic context (e.g., sequence capture and transcriptomics) have remained popular due to their low cost. Second, the independent loci assumptions dramatically simplify the theoretical framework for inferring species trees. Properties of gene trees under the bag-of-genes MSC model are much easier to analyze than models with recombination. Considering gene trees as conditionally independent leads to clean likelihood formulations [52]. Thus, mathematical convenience rather than biological realism has also motivated this choice. While recent progress has been made for calculations under positional models (e.g., the multi-species sequentially Markov coalescent [51]), many questions remain unanswered (see Outstanding questions). Luckily, these models enable simulations that create dependencies across loci, which can be used to evaluate the impact of dependencies and the accuracy of position-aware methods [53,54].

The foundations for methods that treat gene trees as independent realizations of the MSC process already existed in the pioneering work of Pamilo and Nei [55]. Various scalable methods

to analyze gene trees have been designed since then, including those that provide statistical consistency guarantees (Box 2). The scalable ‘gene tree summary’ methods [18] (Box 2) assume that the gene trees used as input are distinct realizations of the MSC model and use expectations under that model. Other methods directly use the likelihood of gene trees given a species tree [56,57], and some additionally use the likelihood of sequences given gene trees [58,59], all using the same MSC model [60]. However, these methods still assume that gene trees are conditionally independent given the species tree, essentially adopting the bag-of-genes model.

Do we lose much by adopting the independence assumption? There have been debates and empirical studies [53,61–65] asking whether recombination negatively affects species tree inference, with several authors [61,64,65] concluding that, under most conditions, it does not (see Rannala [66], for review). Nevertheless, the number of these studies remains limited, and their results are nuanced. Wang and Liu [64] found that recombination has a modest impact on species tree accuracy when they used model trees similar to Lanier and Knowles [61], but they identified other conditions where recombination had a much larger impact. He *et al.* [53] reproduced the results of Wang and Liu [64] in similar parts of the parameter space, but they focused on the nonindependence of adjacent gene trees and concluded that nonindependence was largely problematic only at very low recombination rates. In fact, they further concluded that removing gene trees to reduce the impact of nonindependence almost always decreased the accuracy of ASTRAL, a result that [54] also observed. Furthermore, those studies focused on inference of the species tree topology; other parameters of interest, including measures of uncertainty, may be less robust to errors caused by intralocus recombination. Zhu *et al.* [65] examined the impact of recombination on a more diverse set of parameters, including divergence times, population sizes, introgression probabilities and times, and even species limits, and they found that realistic recombination rates had little impact on parameter estimates. Nevertheless, the parameter space for models that include recombination and mutation is very large, suggesting that additional studies are warranted. Moreover, gene dependencies can lead to double-counting of the same coalescence history, thereby artificially inflating support.

These simulation studies raise two questions. First, why are parameter estimates relatively robust to intralocus recombination? Two neighboring c-genes can differ only by a single subtree-prune-and-regraft (SPR) operation (Box 2) [34,67] and thus will share many features, including the topology of the vast majority of quartets (which matter to many inference methods; see Box 2). Obviously, longer genomic regions may contain more than two c-genes, but the c-genes in a short region are still expected to be associated with relatively similar topologies. While simulation studies have shown that recombination can degrade the performance of standard phylogenetic methods for gene tree estimation [68,69], MSC inference is based on distributions of gene trees. Thus, if the recombinant gene tree distributions retain many of the features of the pure c-genes (e.g., the dominant quartet tree matches the species tree; see anomaly zone in Box 2), these methods can still infer the correct species tree. Nevertheless, this robustness is not total, as simulations by Wang and Liu [64] have shown.

There is a second question that is arguably more important: what do we lose by making the independence assumption? The loss of information might be modest if the only parameter of interest were the species tree. However, there are many other parameters of interest in phylogenomic studies. For example, two studies report evidence that ancestral Neoaves—the clade comprising most extant bird species—had much larger effective population sizes early in their history [70,71], a pattern consistent with ecological release after the mass extinction of dinosaurs (and other taxa). However, those deep-time demographic inferences depend on the independence assumption. The conclusion of Zhu *et al.* [65] that many parameters are robust to violations of the independence assumption may not be general, raising questions about these biological interpretations. Recombination landscapes can be complex, with regions near telomeres and smaller

chromosomes often exhibiting higher recombination rates [22]. Such nonrandom patterns of recombination change may impact downstream analyses, especially when they interact with other processes such as GC-biased gene conversion [72], which can impact gene tree inference [73]. Population processes also have the potential to identify the genes underlying phenotypic changes that occurred in deep time, arguably one of the ‘holy grails’ of evolutionary biology. For example, Feng *et al.* [74] identified candidate genes for morphological change in ancestral marsupials by correlating apparent trait homoplasy with ILS at the underlying loci. It is critical to determine whether such intriguing inferences about events that occurred tens or even hundreds of millions of years ago remain robust when we move beyond the bag-of-genes model.

Practical recombination-aware methods

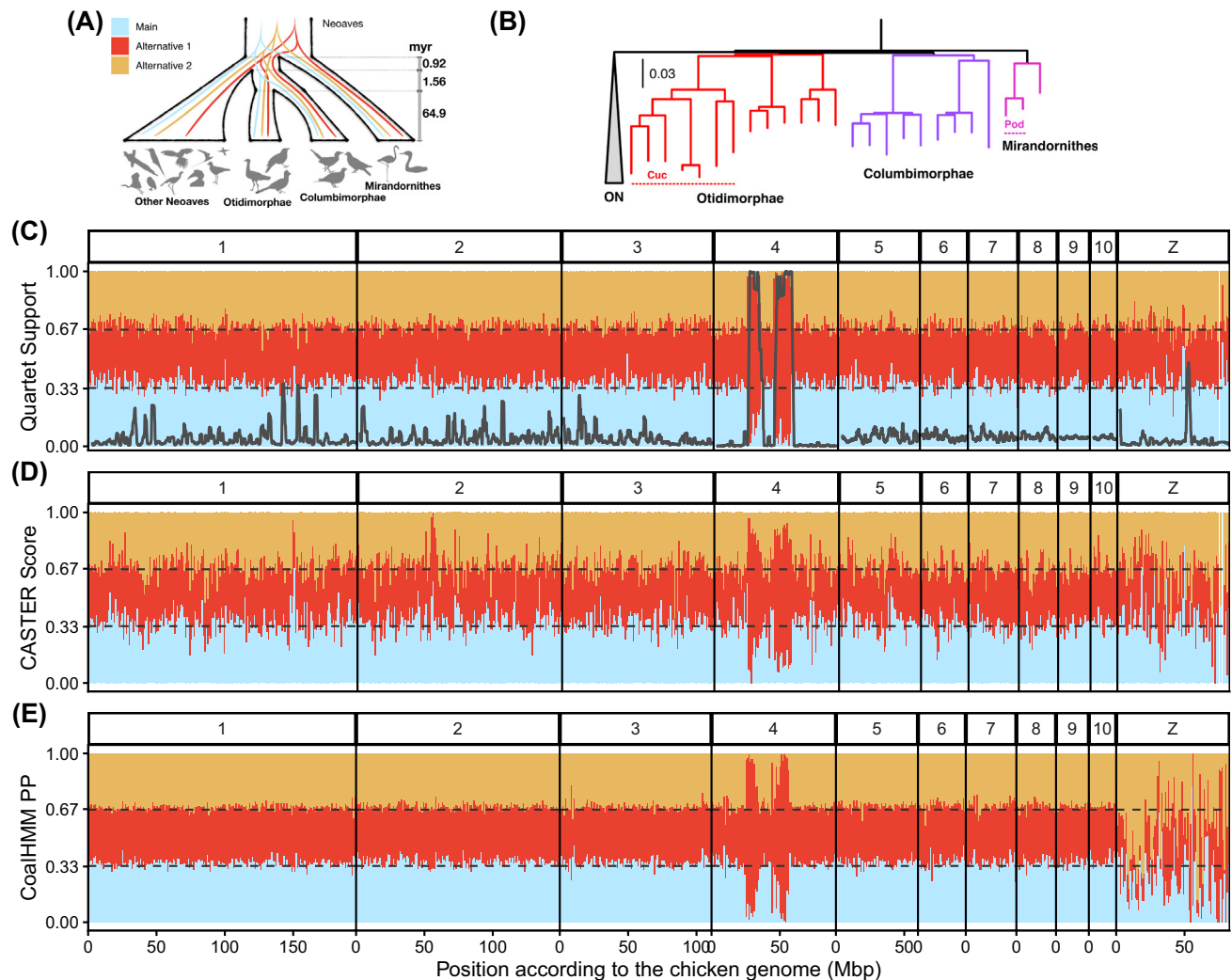
In the absence of ARGs, several approaches enable us to move beyond the bag-of-genes model and incorporate positional information. Below, we outline four categories of these methods (Table 1). These methods differ to some extent in their input, output, and goals. Nevertheless, they can also be used to answer some of the same underlying questions: How do gene trees change as one moves along chromosomes? Do different chromosomes or certain chromosomal

Table 1. Example software tools for position-aware phylogenomics

Approach	Software	Goal	Scalability	Resolution	Input
Post hoc	TreeHouseExp. [75]	Visualization of positional topologies across the genome.	Linear in k ; n in hundreds	Loci used to get gene trees, often short windows (e.g., 1 Kbp)	An ordered set of gene trees
	QQS [14]	Examine moving average of quadripartition quartet score (QQS) per branch.	Linear in k ; quadratic with n		
	Phlag [76]	Uses HMMs to segment genomes to find non-ILS signal per branch.	Linear in k ; quadratic with n		
Site-based	SVDquartets [77]	To infer species trees without demarcating c-genes.	Linear in L ; n in hundreds	Sites	SNP matrix, set of orthologs, multiple genome alignment
	SplitsScores [78]	Similar to SVDquartets but allows examination of the local signal for windows.	10 s for $n = 100$; $L = 10$ Kbp	Sites chunked to windows	
	CASTER [54]	Site-based species tree inference. Site scores can be averages across windows to examine the signal. Scales to large numbers of full genomes.	Linear in L ; approximately quadratic in n	Sites	
Breakpoint detection	PHI [79]	A simple statistical test to find evidence of recombination within a locus.	$n < 30$	Sites, often in windows of 100 bp to tens of Kbp	Multiple genome alignment
	3SEQ [80]	Scalable statistical test for recombination using triples.	n in hundreds; L in tens of Kbp		
	GARD [81]	A tree-based approach using genetic algorithms to scale.	n in tens; L in Kbps		
	RDP [82]	Uses several tests applied on sliding windows using some phylogeny-aware algorithms to detect/visualize recombinations; used most often for viruses.	$n < 5000$; $L < 50$ Mbp		
	biomc2 [83]	Bayesian model comparison under the coalescent-with-recombination model; the full likelihood calculations make it less scalable.	$n < 64$; $L < 10$ Kbp		
Coalescent HMMs	CoalHMM [84]	Estimates speciation times and (biased) ancestral N from alignments of three species. It can infer gene tree topologies and their deep coalescence at the base-pair level.	$n \leq 5$ (often 3 + outgroup); computationally heavy (multithreading, large memory)	Sites	Multiple genome alignments, ideally with large syntenic blocks
	TRAILS [11]	An extension of CoalHMM, which infers coalescent times in discretized time intervals. The demographic inference is unbiased; the posterior decoding has provided per-base-pair temporal information.			
	PhyloNet-HMM [13]	An extension of CoalHMM that allows hybridization.			

Here, n refers to the number of species (i.e., sequences), k refers to the number of gene trees, and L refers to the total sequence length.

regions have gene tree distributions that differ substantially from the rest of the genome? Examples of such outlier regions abound in the literature (e.g., [10–13,15,54]) and are attributed to several causes. We will use the outlier region identified by Mirarab *et al.* [14] to demonstrate how different methods can detect the same signal, albeit with differing inputs, outputs, and interpretative power. This example concerns a region of chromosome 4 in birds and was chosen because it shows a particularly strong signal of reduced recombination, evident from the high frequency of a topology that differs from the species tree for branches dated 65–68 million years ago (Figure 2).



Trends in Genetics

Figure 2. Two outlier regions in the avian genome revealed through three approaches. (A) The species tree inferred by Stiller *et al.* [71]. Differed from earlier analyses by Jarvis *et al.* [73]. In the resolution of the first branching event among Neoaves; Jarvis *et al.* [73], Recovered Alternative 1 (Columbea). (B) Branch lengths for the focal clades are in substitution units, estimated by CASTLES [85]. The very short internal branches at the base of Neoaves are consistent with the high degrees of conflict among gene trees. Dashed lines emphasize the avian orders Cuculiformes (Cuc) and Podicipediformes (Pod), which are characterized by accelerated molecular evolution relative to their sister orders, showing strong deviation from ultrametricity. (C–E) Evidence for ‘outlier regions’ in the bird genome where segments of chromosome 4 have gene trees that strongly support Columbea based on (C) *post hoc* quartet support, (D) site-based CASTER scores, and (E) CoalHMM posterior probabilities. The ‘main’ topology is observed slightly more than 1/3 of the time across most of the chromosome; in a specific region of chromosome 4, the ‘Columbea’ alternative is far more prevalent. Ignoring genomic positions can mislead species tree inference by counting this apparently linked history many times, favoring Columbea. All positions are by chicken; synteny changes across the tree, putting the two outlier regions together in many Neoaves [14]. In panel (c), the gray line shows the posterior decoding of being in an ‘outlier’ state in the Phlag HMM.

Ad hoc and post hoc approaches

Most phylogenomic analyses first infer gene trees for (presumed) c-genes. When each locus is a contiguous genomic region, we can incorporate positional information to examine the recombination landscape. These analyses can be *ad hoc* or *post hoc*, depending on whether they reflect *a priori* decisions to exclude certain types of data (e.g., gene trees based on sex chromosome sequences) or *a posteriori* efforts to understand the distribution of phylogenetic signals on chromosomes. These approaches often use one of the species as a reference and assume full synteny across the tree. The gene trees are then mapped to positions on ‘the genome’, effectively ordering genes.

With genes ordered, we can examine how topologies change as one moves from one end of a chromosome to the other, looking for regional patterns. This can be done visually, and tools that assist in such visualizations exist, including ‘Tree House Explorer’ [75]. However, the number of possible gene tree topologies is large, making them harder to enumerate for larger trees. To address this, one can compute summary statistics that can then be visualized as a signal across the genome. For example, one can look at overall gene tree–species tree discordance using metrics such as the tree distance along the genome as a proxy for recombination rates (e.g., [71]). Or, one can focus on specific species tree branches and visualize the presence of specific bipartitions [16], frequencies of rooted triplets (e.g., [15]), or unrooted quartets (e.g., [14]). Triplets and quartets have nice properties under MSC (Box 2) and enjoy the added benefit that for a specific branch of the species tree, one can efficiently compute a summarized score between 0 and 1, averaging over all triplets/quartets relevant to that branch [86]. This number is intimately connected to branch lengths in coalescent units and is expected to be, on average, above 1/3 for the topology matching the species tree. Strong deviations from the genome-wide average (Figure 2C) can indicate violations of MSC. Thus, these signals can highlight deviations from the null hypothesis that observed discordance can be explained by ILS alone, provide evidence for introgression, and reveal the structure of the recombination landscape for different chromosomes. Recently, a method called Phlag was developed to automatically detect such deviations using HMMs and flag regions inconsistent with ILS-only deviations [76].

Although these success stories show the promise of *post hoc* analyses, their scalability comes at a cost. The current *post hoc* methods do not provide a principled mechanism for updating the topology, branch lengths, or the statistical support for the species tree (see ‘Open questions’). The more fundamental issues with *post hoc* methods are (i) the need to predefine the loci and the distinct possibility that each locus is still a mosaic of histories due to recombination and (ii) the independent inference of locus trees is less statistically efficient than considering all the data jointly.

Site-based methods

An approach that avoids the need to divide sites into loci is site-based species tree estimation. In this approach, we score a species tree with respect to the entire alignment of all sites. Site-based methods do not directly consider site positions and have been suggested to be relatively robust to incorrect c-gene delimitation (e.g., [87]). Essentially, they are viewed as making fewer assumptions by adopting a ‘bag-of-sites’ model instead of the bag-of-genes model.

How are site-based methods different from the long-used supermatrix approach? They are similar in that they concatenate all sites but differ in that they account for coalescence. Simply using maximum likelihood (ML) without considering changes across the genome is not statistically consistent under the MSC [88]. To solve this, one can average the site-pattern probabilities over the density of coalescent times and gene tree topologies under the MSC, an approach

that Zhu and Yang [89] call ‘independent-sites ML’ (ISML) or ‘coalescent-aware concatenation’. However, ISML is currently limited to simple models of sequence evolution, such as Jukes and Cantor [90], and this limitation may be difficult to overcome. Other ways of extending ML to include mixtures of trees [91] or polymorphism-aware models [92] introduce additional complexities, although they warrant further study.

Besides less scalable ISML, methods that use analytical expectations on quartet site patterns under the MSC have also been developed. An early method was SVDquartets [77], a phylogenetic invariants method based on the analytical result that the rank of a ‘flattening matrix’ with the frequencies of 256 possible DNA site patterns should be less than or equal to ten for the correct quartet topology. One can then apply this criterion to some or all quartets and then combine quartet trees [77,93]. While applying SVDquartets to whole-genome alignments is computationally challenging, a sample of sites can be used, an approach successfully employed by Foley *et al.* [16]. A more recent method, CASTER [54], also uses quartet site patterns but instead defines a score that is statistically expected to be maximized locally by each gene tree for each c-gene and globally by the species tree across the genome, summed over all sites and quartets. A variant of CASTER, called CASTER-pair, computes scores over pairs of sites rather than individual sites, enabling inference under more complex models. Because the expectation sums over all quartets, CASTER does not need to compute a tree for each quartet separately; instead, it can efficiently compute the total score without enumerating all quartets, enabling it to scale to 1.8 billion sites from 241 species [54]. CASTER also accounts for site positions when computing support by adopting a block bootstrap approach.

Site-based methods can also be used to interrogate the local signal across the genome. For CASTER, its score is a consistent estimator of the local gene tree [54]. Thus, to incorporate positions, one can compute the average of these scores over sliding windows to detect alignment errors, evidence of introgression, and regions with unusual histories (Figure 1D). Allman *et al.* [78] developed a method called SplitsScores, similar to SVDquartets but extended to more than four taxa, that can be used in a sliding window framework to examine the distribution of phylogenetic signal across the genome, with similar potential.

Breakpoint detection methods

Methods that detect tree-changing recombination breakpoints (Table 1) have the potential to address concerns about the existence of multiple c-genes in alignments used to infer gene trees. These methods can be divided into two classes: (i) tests to determine whether an alignment shows evidence of a recombinant history (relatively fast) and (ii) methods to identify recombination breakpoints that delimit c-genes. Although developed mainly to detect recombinant viruses, recombination detection methods can also be used to identify highly recombined gene trees [94,95] to exclude and boost species tree inference pipelines. Zhao *et al.* [96] used logistic regression to show that a nonparametric metric called 3SEQ [80] (Table 1) was correlated with gene tree error and used this to justify excluding specific gene trees. The c-genes in any aligned region can be delimited using breakpoint detection methods and used to infer gene trees (e.g., [64]). This approach raises a vexing question: which c-genes should be included in species tree inference? The topological similarities between neighboring gene trees are in conflict with MSC, meaning that to be fully compliant with the bag-of-genes model, selected c-genes for species tree inference should be spaced, but the ideal distance between c-genes is unclear. Another challenge is that gene tree topologies are correlated with c-gene lengths [27,51], so simple strategies, such as selecting the longest c-gene in an aligned region to increase phylogenetic signal, may produce a biased set of gene tree topologies. To fully conform to the bag-of-genes MSC model, c-genes will have to be selected only on the basis of their spacing, not on the basis of

length. However, this would lead to the inclusion of some very short c-genes, inflating gene tree estimation error. Furthermore, simulations that include recombination do not show a benefit in removing neighboring loci [53,54]. Fundamentally, even if informed by inferred recombination breakpoints, the bag of c-genes is an imperfect model of the evolutionary history—a position-aware model such as a multi-species ARG is a better representation of genomic history.

Coalescent hidden Markov models

The most direct approach to jointly model coalescence and recombination in a phylogenetic context is the coalescent HMM. In this framework, hidden states represent gene tree topologies and coalescent times, and transitions between states capture recombination breakpoints. These models allow temporal restrictions on when lineages coalesce; in effect, they overlay the species tree on top of the coalescent-with-recombination process to prevent coalescent events post-speciation (but see Liu *et al.* [13], for an extension that also allows gene flow). The simplest of such models considers one sample from each of two species and forbids coalescence before the species split time [97]. This model can also be used to infer demographic parameters, as the parameters contain information about split times and ancestral effective population sizes (N_e), which is inversely related to the rate of coalescence. CoalHMM [84,98] extends this approach by including a third species, allowing it to explicitly model ILS. CoalHMM takes advantage of the genome-wide distribution of discordant genealogies for demographic inference, and it has been used to infer speciation times and ancestral N_e in primates [99–101], birds [71], and marsupials [74]. CoalHMM also infers the multi-species ARG at the base-pair level through posterior decoding, and this can be used to provide position-aware measures of statistical support, detect selection at specific genomic locations [99], reconcile phenotypic incongruence [74], and identify regions with unusual genealogical histories [14] (Figure 2E). An extension of CoalHMM, named TRAILS [11], infers coalescent times in discretized time intervals, enabling more accurate estimation of demographic parameters and a more detailed ARG reconstruction, with topological and time-resolved coalescences.

While coalescent HMMs are useful tools for explicitly modeling the multi-species ARG, they suffer from several important limitations. In particular, these methods are typically restricted to a small number of species and coarse temporal resolution because the complexity of the underlying coalescent-with-recombination framework increases rapidly with both sample size and the number of discretized time intervals. By contrast, a range of approximations developed for within-population ARG reconstruction (e.g., [44–48]) enable analyses of hundreds to thousands of samples but have not yet been extended to a multi-species setting. This gap largely reflects additional challenges inherent to multi-species data, such as heterogeneity in mutation rates (leading to violations of ultrametricity), violations of the infinite-sites model (necessitating the use of more complex models of sequence evolution), and the inclusion of barriers to gene flow that reflect speciation.

What are the best ways to obtain positional information?

Inference based on full multi-species ARGs requires highly contiguous and accurate sequence alignments. However, most available genomes remain fragmented, limiting the ability to reliably assign sequences to genomic positions. Even with less-than-ideal genomes, positional information alone can still be informative, as illustrated by our descriptions of *ad hoc* methods applied to relatively low-quality genome assemblies (e.g., [14], shown in Figure 2). These challenges are exacerbated for legacy datasets that do not rely on genome assemblies but instead consist of ortholog sets, such as coding sequences (e.g., from transcriptome data) or Ultra-Conserved Elements (UCEs). Because position-aware phylogenomics rests on the concept of positional orthology [102], applying these approaches to legacy data requires

mapping sequences onto high-quality reference genomes and, in doing so, assuming positional orthology. This assumption may be reasonable for contiguous elements such as UCEs, but it breaks down for multi-exon genes [62], where distantly separated genomic regions may be incorrectly treated as contiguous. A workaround is to decompose the assembled transcripts into individual exons by mapping them to reference genomes. Still, synteny may shift over deep time, and short exons may be difficult to use for gene tree inference. Furthermore, positional analyses of legacy data will be challenging because marker spacing is uneven relative to that achievable for regions extracted from whole-genome alignments, thereby affecting the resolution of the positional signal. Overall, phylogenomic analyses that incorporate positional information are best suited to whole-genome data, with positional analyses of legacy datasets providing only coarse resolution.

A separate set of challenges arises from reduced synteny over deep time and/or the inclusion of taxa with high rates of structural change due to genome rearrangements [103] (see Outstanding questions). While changes in synteny can provide a strong phylogenetic signal, a topic that has been extensively studied (including with phylogenomic datasets [104]), it complicates assigning co-linear positions. One could attempt to conduct analyses within a framework that includes ancestral state reconstructions for genome structures, but this is impractical because models for combined analyses that jointly consider substitutions, recombination, and synteny changes are not currently available. At this point in time, position-aware phylogenomics will have to rely on heuristics to address these practical challenges. Researchers will also have to accept that imperfect conservation of synteny will be a source of error and that specific heuristics may fail at deep time scales. However, it is clear that simple approaches, such as using a reference genome to define coordinates, can provide substantial insights [14,76]. Moving beyond these approximations will require developing novel statistical models and computational methods to integrate information across multiple sources.

Concluding remarks and future perspectives

As higher quality genomes, including multiple haplotype-resolved genomes, become increasingly available, phylogenomics can shift from a bag-of-genes model to a tree-sequence approach, more closely resembling the population genetics literature. Methods have emerged to enable this new model, and the benefits of the paradigm shift have been increasingly documented. The math for modeling tree-sequences under a multi-species model is more complex than either tree-sequences for a single population or bag-of-genes for multi-species. Thus, this aspiration must contend with computational challenges, a topic we anticipate will generate significant interest in the near future. More broadly, while species tree inference may benefit from a tree-sequence model, we believe the main benefits lie in inferring other parameters. Given a species tree, positional-aware methods (whether *post hoc*, site-based, or HMM-based) could look for non-ILS signals, including but not limited to hybridization, strong selection, artifacts due to model misspecification (long-branch attraction, heterotachy), and artifacts due to data errors such as contamination or homology errors. They can also attempt to detect changes in ILS parameters, including population size and recombination rates. These inferences will, in turn, be helpful for attempts to understand how genomes have evolved and how they produce phenotypes observed among living species [105].

Acknowledgments

This work is supported by the National Institutes of Health (1R35GM142725).

Declaration of interests

The authors declare no competing interests.

Outstanding questions

Is it possible to develop fast and accurate methods to identify recombination breakpoints and thus define *coalescence genes* at the phylogenomic scale, perhaps adopting the machinery used in site-based methods?

Can scalable gene tree summary methods or site-based methods, which are agnostic to gene/site positions, be updated to account for genomic position, perhaps by accounting for autocorrelation using weighting strategies?

How can within-population ancestral recombination graph reconstruction methods (e.g., ARGweaver and Relate) be extended to multiple species? How can the already complex coalescent-with-recombination framework integrate the species tree (i.e., explicit modeling of barriers to gene flow) and more complex sequence-evolution models, including those that account for lineage-specific rate variations prevalent at phylogenetic depths, while maintaining scalability?

How do recombinations interact with genome rearrangements or imperfect assemblies? How do we mathematically model the combined effects of recombinations and rearrangements to enable position-aware analyses that account for reduced synteny? How do we model periods of time when genome rearrangements remain polymorphisms over multiple speciation events or when rearrangements accelerate?

Do position-aware phylogenomic methods need new ways of measuring statistical support? A lack of independence may make methods such as block bootstrapping more appropriate, but how exactly should they be adopted?

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